

Hypothesis Testing and Statistical Significance

Defensible tests, uncertainty, false discoveries, effect sizes, and research decisions.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

INTERMEDIATE

1. BEGIN WITH THE DECISION

State the action, population, estimand, and minimum useful effect before choosing a test. The statistical design should protect the decision from its most costly error.

2. SEPARATE EXPLORATION FROM CONFIRMATION

Explore patterns freely, but log every choice and reserve untouched evidence for fixed claims. Once outcomes change a decision, those outcomes are no longer a holdout.

3. MATCH UNCERTAINTY TO STRUCTURE

Identify independent units, overlap, clusters, weights, and regimes. Standard errors and resampling must reproduce that structure rather than treating every database row as new evidence.

4. READ MAGNITUDE BEFORE THRESHOLDS

Inspect raw effect, interval, practical hurdle, and cost stress before the p-value. A threshold crossing is a calibrated decision event, not a complete interpretation.

5. ACCOUNT FOR SEARCH

Define the hypothesis family, record failed variants, and adjust or confirm independently. The final model inherits the full path that selected it.

6. CHALLENGE THE MECHANISM

Use diagnostics, robust estimators, placebo designs, subgroup interactions, and temporal replay. A result should survive the failure modes its deployment will face.

READING PATTERN

Each concept page moves from meaning to memory cue, quant use, formula intuition, market example, implementation, failure mode, system placement, and active recall.

Notation and inference contract

INTERMEDIATE

SYMBOL	OBJECT	MEANING
theta	estimand	Population quantity the study intends to learn
theta_hat	estimate	Value computed from the observed eligible sample
theta_0	null value	Reference value specified by the null hypothesis
SE	standard error	Estimated sampling variability of theta_hat
T	test statistic	Standardized incompatibility with the null
p	p-value	Null tail area at least as extreme as observed
alpha	test level	Predeclared Type I error budget
beta	Type II rate	Miss probability at a specified alternative
1-beta	power	Detection probability at that alternative
Delta	practical margin	Minimum useful or acceptable effect boundary

ALWAYS ATTACH

Population, grain, cutoff, horizon, outcome, contrast, units, weights, exclusions, hypothesis side, search family, uncertainty method, cluster or block rule, effect hurdle, and decision action.

INTERPRETATION BOUNDARY

The estimate describes magnitude, the interval describes compatible values under a procedure, the p-value calibrates null surprise, and the gate decides action. None can substitute for the others.

Hypothesis-testing research workflow

INTERMEDIATE

1

Frame the decision

Action, population, estimand, cost of errors, useful effect

2

Register the design

Null, alternative, side, family, data, estimator, stopping

3

Build the sample

Point-in-time eligibility, grain, missingness, clusters, horizon

4

Inspect assumptions

Influence, dependence, regimes, overlap, variance, support

5

Estimate magnitude

Raw effect, practical scale, costs, contribution, stability

6

Calibrate uncertainty

Analytic, robust, cluster, permutation, or bootstrap method

7

Correct and challenge

Multiplicity, holdout, placebo, specification, stress, replay

8

Decide and monitor

Gate, artifact, reviewer, expiry, pilot, drift, rollback

HARD GATE

No claim reaches capital allocation until the estimand, search path, independent information, uncertainty method, practical hurdle, and holdout status are reproducible from a sealed artifact.

Research question, estimand, and evidence

INTERMEDIATE

ONE-SENTENCE MEANING

A research question becomes testable only after the estimand states the exact population quantity the study intends to learn. Evidence is then judged against that target, not against a vague trading story.

MEMORY JOGGER

Finish the sentence: among which eligible observations, what contrast at what horizon do I want to estimate? If that sentence changes, the hypothesis and required data usually change with it.

WHY IT MATTERS IN QUANT RESEARCH

Quant studies often mix signal prediction, portfolio payoff, and implementation quality in one claim. Separating their estimands keeps a statistically convincing feature result from being mistaken for a deployable strategy result.

FORMULA INTUITION

An estimand can be written $\theta = E[Y(1) - Y(0)]$ or $\theta = E[r \mid \text{signal high}] - E[r \mid \text{signal low}]$. The notation is useful only when population, timing, weighting, and treatment or grouping rules are explicit.

SIMPLE MARKET EXAMPLE

A momentum study may estimate the next-month return spread between top and bottom deciles in liquid US equities. That estimand differs from the after-cost return of a capacity-constrained long-short portfolio.

PYTHON / SYSTEM IMPLEMENTATION

Store a machine-readable study contract with universe, cutoff, horizon, outcome, contrast, weights, exclusions, and unit. Generate the analysis table from that contract so notebooks cannot silently redefine the target.

CONFUSIONS TO AVOID

Do not begin with a favored test and search for a compatible question. A low p-value for the wrong estimand is precise evidence about something the strategy does not need.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research registry binds every hypothesis to an estimand ID before data extraction. Validation, backtesting, and monitoring reuse that ID or document why deployment requires a different target.

RELATED CONCEPTS AND QUICK RECALL

Related: sampling frame, causal contrast, prediction target, economic objective. Recall task: Write separate estimands for predictive accuracy, gross alpha, and after-cost portfolio value.

Null and alternative hypotheses

INTERMEDIATE

ONE-SENTENCE MEANING

The null hypothesis defines the reference claim the data must challenge, while the alternative states the scientifically relevant departure. Together they determine what outcomes count as evidence and what the test cannot establish.

MEMORY JOGGER

The null is a benchmark model, not a claim that nature literally equals zero. The alternative should encode direction or range only when the research question justified it before seeing results.

WHY IT MATTERS IN QUANT RESEARCH

A clean hypothesis prevents researchers from interpreting any favorable statistic as support. It also makes clear whether the study concerns mean return, rank information, turnover reduction, drawdown control, or another distinct object.

FORMULA INTUITION

For a mean spread, $H_0: \mu_{\text{spread}} = 0$ and $H_1: \mu_{\text{spread}} \neq 0$ is two-sided. A superiority claim may use $H_1: \mu_{\text{spread}} > 0$, but the sign must be fixed before inspection.

SIMPLE MARKET EXAMPLE

Testing whether a new execution model changes slippage uses a two-sided alternative. Testing whether it reduces slippage may be one-sided only when an increase would never be relabeled as a different success.

PYTHON / SYSTEM IMPLEMENTATION

Represent hypotheses as structured fields for parameter, null value, operator, side, and practical threshold. Unit tests should reject analysis code whose displayed hypothesis differs from the registered specification.

CONFUSIONS TO AVOID

Failing to reject H_0 does not prove H_0 , and rejecting H_0 does not prove the alternative mechanism. Composite nulls, nuisance parameters, and model misspecification can all complicate the simple wording.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment object stores both statistical and economic hypotheses. Promotion logic requires the observed interval to address each one rather than treating rejection alone as a deployment verdict.

RELATED CONCEPTS AND QUICK RECALL

Related: reference model, superiority, equivalence, composite hypothesis. Recall task: State null and alternative hypotheses for a signal whose minimum useful net return is 15 basis points monthly.

Test statistic and reference distribution

INTERMEDIATE

ONE-SENTENCE MEANING

A test statistic compresses sample evidence into a quantity whose behavior is known or approximated under the null. The reference distribution converts that quantity into tail probability or a critical boundary.

MEMORY JOGGER

Evidence is not the raw estimate alone; it is the estimate measured against expected sampling noise. Different null mechanisms can assign different meanings to the same observed spread.

WHY IT MATTERS IN QUANT RESEARCH

Financial observations are dependent, heteroskedastic, and often selected, so textbook reference distributions can be badly calibrated. Choosing the statistic and null generator is therefore a modeling decision, not a formatting step.

FORMULA INTUITION

A common statistic is $T = (\hat{\theta} - \theta_0) / \text{SE}(\hat{\theta})$. Under suitable conditions T may follow a t distribution, converge to normality, or be compared with a permutation distribution.

SIMPLE MARKET EXAMPLE

A 20 bp monthly spread with 5 bp standard error gives $T=4$, while the same spread with 15 bp error gives $T=1.33$. Economic estimate is unchanged, but evidence strength is not.

PYTHON / SYSTEM IMPLEMENTATION

Implement statistic calculation separately from standard-error or resampling logic. Save the observed statistic, every resampled statistic when feasible, degrees of freedom, and the exact reference method.

CONFUSIONS TO AVOID

A large statistic is not automatically valid if its denominator ignores serial dependence or clustering. A normal curve drawn over a distorted null does not repair calibration.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The inference service accepts a sealed analysis table and emits estimate, uncertainty, statistic, reference distribution, and diagnostics as one versioned artifact. Downstream dashboards never recompute only the p -value.

RELATED CONCEPTS AND QUICK RECALL

Related: standard error, critical value, pivotal statistic, null simulation. Recall task: Explain how changing only the null generator can change the interpretation of an identical test statistic.

P-values: meaning and limits

INTERMEDIATE

ONE-SENTENCE MEANING

A p-value is the probability, under the null model, of obtaining a statistic at least as incompatible with the null as the observed one. It is not the probability that the null hypothesis is true.

MEMORY JOGGER

Read p as a tail-area calibration conditional on assumptions and the analysis plan. It measures surprise under a reference process, not strategy quality, repeatability, or expected profit.

WHY IT MATTERS IN QUANT RESEARCH

P-values can flag results worth deeper investigation, but capital decisions need effect size, uncertainty, costs, multiplicity, and robustness. In large samples, immaterial effects can be highly significant.

FORMULA INTUITION

For a two-sided symmetric test, $p = 2 P_0(T \geq |t_{\text{obs}}|)$. Exact formulas vary with the statistic, side, discrete support, and whether nuisance parameters were estimated.

SIMPLE MARKET EXAMPLE

A p-value of 0.03 means that null-compatible repetitions would produce evidence this extreme or more about 3% of the time. It does not mean the strategy has a 97% chance of working.

PYTHON / SYSTEM IMPLEMENTATION

Return p-values with full precision internally and sensible precision in reports. Store sidedness, null value, correction family, and code version so 0.049 cannot be detached from its design.

CONFUSIONS TO AVOID

Do not rank strategies by p-value alone, and do not treat 0.049 and 0.051 as qualitatively different truths. Selective reporting and repeated peeking make nominal p-values misleading.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence catalog displays p beside estimate, interval, sample structure, and correction. Automated decisions use registered thresholds and never parse rounded text from a report.

RELATED CONCEPTS AND QUICK RECALL

Related: tail probability, calibration, confidence interval, selective inference. Recall task: Translate $p=0.08$ into correct language without saying the null is likely true or the signal is useless.

Significance level and decision rule

ONE-SENTENCE MEANING

The significance level α sets the long-run false-rejection rate promised by a valid test under the null. A decision rule maps the statistic or p-value to reject or not reject before results are observed.

MEMORY JOGGER

Alpha is a risk budget for erroneous claims, not a universal law of 5 percent. The acceptable level should reflect search scale, decision cost, reversibility, and evidence stage.

WHY IT MATTERS IN QUANT RESEARCH

Exploratory screening, model promotion, and capital expansion deserve different thresholds. A single alpha across thousands of ideas can generate a large queue of false discoveries.

FORMULA INTUITION

Reject H_0 when $p \leq \alpha$, or equivalently when T enters the precomputed critical region. This equivalence requires the same sidedness and reference distribution in both calculations.

SIMPLE MARKET EXAMPLE

At $\alpha=0.05$, 1,000 independent true-null tests produce about 50 false rejections on average. The realized count varies, and dependence among tests changes its distribution.

PYTHON / SYSTEM IMPLEMENTATION

Configure alpha by study stage in policy rather than notebook constants. Reports should show the threshold, correction method, and whether the result crossed statistical and economic gates.

CONFUSIONS TO AVOID

A result above alpha is not evidence of no effect, while a result below alpha is not a guarantee of replication. Moving alpha after seeing the p-value invalidates the stated error rate.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The governance layer defines exploratory, confirmatory, and production evidence tiers. Each tier specifies alpha, multiplicity control, required holdout, and permissible action.

RELATED CONCEPTS AND QUICK RECALL

Related: critical region, decision threshold, false positive, evidence tier. Recall task: Choose different alpha policies for exploratory factor screening and irreversible production rollout, then justify the difference.

Type I and Type II errors

INTERMEDIATE

ONE-SENTENCE MEANING

A Type I error rejects a true null, while a Type II error fails to reject a false null. These errors describe repeated-decision behavior and must be evaluated against the economic consequences of each mistake.

MEMORY JOGGER

False alarm is Type I; missed opportunity is Type II. Lowering one error usually raises the other unless sample information, measurement quality, or design improves.

WHY IT MATTERS IN QUANT RESEARCH

Quant research faces asymmetric costs: false alpha consumes capital and attention, while missed alpha has opportunity cost. Risk controls may rationally demand stricter false-positive protection than idea generation.

FORMULA INTUITION

$P(\text{Type I}) = \alpha$ for a calibrated simple null, and $P(\text{Type II}) = \beta$ at a specified alternative. Power equals $1 - \beta$ and cannot be stated without naming an effect size.

SIMPLE MARKET EXAMPLE

Rejecting a useless signal can cause turnover and drawdown; missing a modest signal may only defer profit. The relative costs should influence sample size, threshold, and validation intensity.

PYTHON / SYSTEM IMPLEMENTATION

Create a decision-cost table before analysis with actions, false-positive loss, false-negative loss, and reversibility. Simulate both errors under realistic dependence and cost assumptions.

CONFUSIONS TO AVOID

Type II error is not $1 - p$, and alpha does not describe the fraction of published claims that are false. Both rates are conditional on a design and data-generating state.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research operating model attaches error costs to promotion gates. High-cost strategies require lower false-discovery tolerance and stronger replication than low-risk monitoring features.

RELATED CONCEPTS AND QUICK RECALL

Related: false discovery, missed detection, decision theory, power. Recall task: Describe Type I and Type II errors for replacing an execution algorithm and identify which is operationally costlier.

Power and minimum detectable effect

INTERMEDIATE

ONE-SENTENCE MEANING

Power is the probability that a valid test rejects the null when a specified alternative is true. Minimum detectable effect is the smallest effect that reaches a target power for a fixed design and alpha.

MEMORY JOGGER

Power asks whether the study can see a useful effect through noise. It is not a retrospective compliment awarded because the observed p-value happened to be small.

WHY IT MATTERS IN QUANT RESEARCH

Underpowered studies produce wide uncertainty, unstable signs, and exaggerated surviving estimates. Power planning forces the researcher to reconcile data horizon with the smallest economically useful edge.

FORMULA INTUITION

For a simple mean test, signal-to-noise grows roughly as $\delta \sqrt{n}/\sigma$. Dependence replaces n with effective information, and multiple-testing correction raises the critical hurdle.

SIMPLE MARKET EXAMPLE

If a strategy needs 20 bp monthly to cover costs but the study can only detect 60 bp reliably, failure to reject says little about viability. More independent periods or lower-variance design are needed.

PYTHON / SYSTEM IMPLEMENTATION

Simulate power using the intended estimator, overlap, clustering, missingness, and realistic return distribution. Save power curves across effect sizes instead of reporting one optimistic number.

CONFUSIONS TO AVOID

Observed post-hoc power is usually a transformation of the p-value and adds little. Planning with IID observations when returns overlap can dramatically overstate detection ability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Study intake requires an MDE tied to economics and a power analysis tied to the actual sampling structure. Insufficiently powered questions remain exploratory and cannot pass confirmatory gates.

RELATED CONCEPTS AND QUICK RECALL

Related: sample size, effective observations, noncentrality, design sensitivity. Recall task: Explain why doubling calendar length may add far less than double the information for overlapping monthly returns.

Effect size and economic materiality

INTERMEDIATE

ONE-SENTENCE MEANING

Effect size quantifies the magnitude of a relationship, while economic materiality asks whether that magnitude changes a real decision after costs and constraints. Statistical significance addresses neither question by itself.

MEMORY JOGGER

Ask how much, in what unit, and relative to what operational threshold. A precisely estimated tiny edge can be scientifically real and economically irrelevant.

WHY IT MATTERS IN QUANT RESEARCH

Trading choices depend on net basis points, capacity, drawdown, turnover, and capital efficiency. Standardized effects help comparison, but raw economic units determine whether implementation is worthwhile.

FORMULA INTUITION

A standardized mean effect is $d = (\mu_1 - \mu_0) / s_{\text{pooled}}$, while a raw spread remains in return units. Net effect subtracts estimated costs and may require a conservative cost uncertainty buffer.

SIMPLE MARKET EXAMPLE

A 2 bp daily improvement can be highly significant across millions of orders yet vanish after queue position and fees. A 30 bp monthly effect with wide uncertainty may still merit a controlled pilot.

PYTHON / SYSTEM IMPLEMENTATION

Report raw, standardized, and net effects with intervals. Connect every promotion gate to a minimum useful value and stress the cost model before classifying the result.

CONFUSIONS TO AVOID

Standardized effects can look large when denominator volatility is artificially low, and annualization can magnify unstable estimates. Statistical precision does not validate the cost or capacity model.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The strategy scorecard separates evidence strength, economic value, and operational feasibility. No single traffic-light status can substitute for these three dimensions.

RELATED CONCEPTS AND QUICK RECALL

Related: practical significance, standardized effect, cost model, capacity. Recall task: Evaluate a precise 3 bp turnover-adjusted edge when deployment requires at least 8 bp to justify operational risk.

One-sided and two-sided tests

INTERMEDIATE

ONE-SENTENCE MEANING

A one-sided test concentrates rejection probability in one direction, while a two-sided test protects against meaningful departures either way. Sidedness is determined by the question and action, not by the observed sign.

MEMORY JOGGER

Use one side only when the opposite direction would never count as success and would trigger a different conclusion. Otherwise reserve error probability for both tails.

WHY IT MATTERS IN QUANT RESEARCH

Direction can be defensible for a pre-specified superiority or noninferiority claim. Most exploratory alpha studies need two-sided testing because sign reversals are informative and post-hoc direction is common.

FORMULA INTUITION

A two-sided test with level alpha uses $\alpha/2$ in each symmetric tail. A one-sided test uses alpha in one tail, giving more power there but no rejection claim in the opposite tail.

SIMPLE MARKET EXAMPLE

A cost-reduction model may use H_1 : mean cost change < 0 if any increase is simply failure. A new return signal should usually allow either positive or negative predictive association.

PYTHON / SYSTEM IMPLEMENTATION

Encode side in the study specification and prohibit runtime switching. Unit tests should compare the displayed hypothesis, critical value, p-value formula, and interval type.

CONFUSIONS TO AVOID

Halving a two-sided p-value after seeing a favorable estimate is invalid. A large result in the untested direction is not harmless; it may reveal damage that the one-sided claim ignored.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry locks sidedness before holdout access and records the business action associated with each tail. Reviewers see any deviation as a new exploratory study, not a small edit.

RELATED CONCEPTS AND QUICK RECALL

Related: directional claim, critical tail, noninferiority, preregistration. Recall task: Decide sidedness for testing whether a hedge reduces drawdown without increasing ordinary-period variance, and explain both components.

Sampling unit and independence

INTERMEDIATE

ONE-SENTENCE MEANING

The sampling unit is the smallest entity treated as an independent information contribution. Independence assumptions govern how uncertainty accumulates and are often violated by repeated assets, shared dates, and overlapping horizons.

MEMORY JOGGER

Count information clusters, not rows. Ten thousand asset-day records may contain only a few hundred date shocks and a limited number of persistent names.

WHY IT MATTERS IN QUANT RESEARCH

Ignoring dependence makes standard errors too small and discoveries too easy. The correct unit depends on the mechanism: market-wide signals cluster by date, execution outcomes may cluster by order parent, and panels can cluster both ways.

FORMULA INTUITION

Under independence $\text{Var}(\text{mean}) = \sigma^2/n$. With intraclass correlation ρ and average cluster size m , a rough design effect is $1 + (m-1)\rho$, reducing effective sample size.

SIMPLE MARKET EXAMPLE

A cross-sectional factor tested on 2,000 stocks for 250 days does not have 500,000 independent observations. Common daily shocks can make date-level information closer to 250 contributions.

PYTHON / SYSTEM IMPLEMENTATION

Assign stable cluster IDs at data construction and carry them into every estimator. Compare naive, one-way, two-way, and block-resampled uncertainty as a design diagnostic.

CONFUSIONS TO AVOID

Clustering is not an automatic robustness sticker; too few clusters can still invalidate asymptotics. Choosing clusters after seeing significance invites another form of specification search.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The analysis table declares observational grain, resampling unit, and uncertainty cluster separately. The inference engine rejects incompatible combinations rather than guessing from column names.

RELATED CONCEPTS AND QUICK RECALL

Related: panel data, design effect, cluster-robust error, block bootstrap. Recall task: Identify the likely independent units in a study of intraday fills nested within orders, symbols, and trading days.

Stationarity, regimes, and comparability

INTERMEDIATE

ONE-SENTENCE MEANING

A hypothesis test assumes observations are comparable under a defined data-generating process or that modeled changes are handled correctly. Regime shifts can turn one pooled parameter into an average of incompatible states.

MEMORY JOGGER

Before testing whether a mean differs from zero, ask whether one stable mean is a coherent object. Time variation can create rejection, cancellation, or false precision depending on state frequencies.

WHY IT MATTERS IN QUANT RESEARCH

Market structure, volatility, liquidity, and policy change over time. Conditional estimates and stability diagnostics often matter more than a single full-history significance result.

FORMULA INTUITION

A pooled effect $\theta = \sum_s \pi_s \theta_s$ depends on regime weights π_s . Changing deployment frequencies can change the relevant effect even when each conditional θ_s is stable.

SIMPLE MARKET EXAMPLE

A strategy earns 40 bp in calm months and loses 80 bp in stress months. A historical average near zero conceals both a useful conditional pattern and a severe tail dependency.

PYTHON / SYSTEM IMPLEMENTATION

Plot rolling estimates with intervals, predefine regime labels, test interactions, and run break diagnostics. Use only states observable at the decision time and report counts for every segment.

CONFUSIONS TO AVOID

Splitting into many ex-post regimes creates multiplicity and tiny samples. A significant full-history average does not guarantee persistence, while an insignificant mixture does not imply no conditional effect.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Validation requires temporal stability panels and explicit deployment-state assumptions. Monitoring compares live state mix and conditional performance with the sealed research artifact.

RELATED CONCEPTS AND QUICK RECALL

Related: structural break, interaction, conditional effect, concept drift. Recall task: Explain how a stable conditional edge can disappear when regime frequencies shift between research and deployment.

Sample size and effective information

INTERMEDIATE

ONE-SENTENCE MEANING

Nominal sample size counts rows, while effective information discounts redundancy from dependence, unequal weights, overlap, and weak variation. Precision follows effective information rather than database volume.

MEMORY JOGGER

Rows are not evidence tokens. Ask how many distinct shocks, dates, entities, and treatment changes actually identify the estimate.

WHY IT MATTERS IN QUANT RESEARCH

Overstated sample size produces narrow intervals and unrealistic power. Effective information also explains why adding more high-frequency records may not improve inference about monthly strategy performance.

FORMULA INTUITION

For a stationary series with autocorrelations ρ_k , n_{eff} is approximately $n/[1+2 \sum_k \rho_k]$ under suitable truncation. Weighted samples can use $n_{\text{eff}} = (\sum w)^2 / \sum w^2$ as a separate concentration diagnostic.

SIMPLE MARKET EXAMPLE

Sixty overlapping 12-month returns from six years do not provide sixty independent annual outcomes. Their shared months create strong positive dependence and far less new information.

PYTHON / SYSTEM IMPLEMENTATION

Report row count, cluster count, date span, effective sample estimate, and sensitivity to lag or block choice. Power simulations should reproduce the same overlap and weighting scheme.

CONFUSIONS TO AVOID

There is no single universal effective sample size for every estimator. Applying an autocorrelation shortcut to a cross-sectional cluster problem can be as misleading as using raw n .

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data-quality summary exposes multiple information counts beside every result. Promotion gates use conservative counts appropriate to the estimator and failure mode.

RELATED CONCEPTS AND QUICK RECALL

Related: autocorrelation, Kish sample size, overlap, information content. Recall task: Contrast nominal and effective sample size for daily observations of a slow-moving valuation factor.

Pre-specification and research degrees of freedom

INTERMEDIATE

ONE-SENTENCE MEANING

Pre-specification records the question, data, transformations, test, and decision rule before confirmatory results are known. It limits silent choices that make favorable evidence easier to obtain.

MEMORY JOGGER

Every adjustable filter, horizon, universe, lag, loss function, and stopping rule is a research degree of freedom. Exploration is legitimate, but confirmation must disclose and separate it.

WHY IT MATTERS IN QUANT RESEARCH

Quant workflows can search thousands of plausible variants without calling the process multiple testing. A precise pre-specified replay gives the holdout result a meaning that a flexible notebook cannot.

FORMULA INTUITION

A specification is a mapping $S(\text{data_version}, \text{cutoff}, \text{transforms}, \text{estimator}, \text{decision_rule})$. Evidence calibration assumes S was not selected using the same outcomes it evaluates.

SIMPLE MARKET EXAMPLE

Choosing a 63-day lookback after comparing 20 alternatives makes the final p-value conditional on a hidden search. Registering the chosen rule before a new period restores a confirmatory test.

PYTHON / SYSTEM IMPLEMENTATION

Serialize study specifications, hash them, and lock confirmatory fields before holdout access. Record all exploratory variants and their results in a search ledger rather than deleting failures.

CONFUSIONS TO AVOID

Pre-specification does not prohibit learning or guarantee a good model. It distinguishes discovery from confirmation and makes deviations visible enough to assess honestly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The research platform enforces separate exploratory and sealed-confirmatory workspaces. Any change after unblinding creates a new study version and consumes new validation data.

RELATED CONCEPTS AND QUICK RECALL

Related: preregistration, specification curve, holdout, audit trail. Recall task: List six degrees of freedom in a momentum backtest that should be frozen before confirmatory evaluation.

Multiple testing and family definition

INTERMEDIATE

ONE-SENTENCE MEANING

Multiple testing arises when many hypotheses are considered and any favorable result may be selected. Error control requires defining the family of claims that compete for the same decision.

MEMORY JOGGER

One reported test can conceal a thousand attempted tests. The relevant multiplicity is the search process, not the number of rows in the final table.

WHY IT MATTERS IN QUANT RESEARCH

Feature libraries, universes, horizons, and parameter grids create broad hypothesis families. Without correction or independent confirmation, the best-looking signal is often a noise winner.

FORMULA INTUITION

Bonferroni controls family-wise error by testing each of m claims at α/m . False discovery rate procedures order p -values and control the expected proportion of false rejections under stated dependence conditions.

SIMPLE MARKET EXAMPLE

With 200 true-null signals at $\alpha=0.05$, about ten false positives are expected before selection. Reporting only the lowest p -value makes that ordinary outcome appear exceptional.

PYTHON / SYSTEM IMPLEMENTATION

Log every attempted hypothesis with a family ID and immutable outcome. Apply the correction to the full family, or reserve an untouched dataset for a new single confirmatory claim.

CONFUSIONS TO AVOID

Bonferroni can be conservative, but using no correction is not automatically more scientific. FDR controls a rate among discoveries, not the probability that any one selected claim is true.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The search service meters a study's hypothesis budget and emits adjusted evidence. Model cards show both raw and family-adjusted values plus the family construction rationale.

RELATED CONCEPTS AND QUICK RECALL

Related: family-wise error, false discovery rate, search ledger, p -hacking. Recall task: Define a defensible testing family for 30 features evaluated across 5 horizons and 4 universes.

Data snooping and holdout integrity

INTERMEDIATE

ONE-SENTENCE MEANING

Data snooping occurs when outcomes influence model choices and the same data later evaluate those choices as if they were fixed. A holdout is informative only while its outcomes remain unconsumed by iteration.

MEMORY JOGGER

A dataset becomes training data the moment it changes a decision. Calling it test data does not preserve independence after repeated viewing.

WHY IT MATTERS IN QUANT RESEARCH

Backtests make leakage inexpensive because researchers can adapt stories, filters, and parameters to full history. Genuine out-of-sample evidence requires process controls, not merely a date split.

FORMULA INTUITION

The reported test conditions on a fixed model M , but snooping selects $M_{\text{hat}} = \text{data-dependent argmax performance}$. The null distribution of the maximum is wider than that of one preselected model.

SIMPLE MARKET EXAMPLE

A team checks the 2025 holdout, changes transaction-cost logic, and checks again. Even a justified change uses information from that period, so the period no longer supports an untouched claim.

PYTHON / SYSTEM IMPLEMENTATION

Use blinded outcome columns, access logs, sealed datasets, and one-way promotion. When a holdout informs any change, relabel it validation data and acquire a new future or external test.

CONFUSIONS TO AVOID

A longer holdout does not survive unlimited iteration, and nested cross-validation does not excuse manual peeking outside its loop. Process leakage can occur without a code-level feature leak.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The platform grants holdout access through auditable jobs that return predefined metrics. Human-visible traces are minimized until a formal unblinding decision is recorded.

RELATED CONCEPTS AND QUICK RECALL

Related: selection bias, adaptive overfitting, nested validation, unblinding. Recall task: Describe when a holdout must be retired after a researcher inspects failure cases and revises the strategy.

Sequential testing and optional stopping

INTERMEDIATE

ONE-SENTENCE MEANING

Sequential testing evaluates accumulating data while preserving a planned error guarantee. Optional stopping without adjustment repeatedly asks a fixed-sample question until chance produces significance.

MEMORY JOGGER

Every peek spends information or error budget. A valid sequential design prices those peeks in advance instead of pretending the final look was the only look.

WHY IT MATTERS IN QUANT RESEARCH

Live experiments and strategy pilots naturally accumulate evidence over time. Fixed-horizon p-values become anti-conservative when teams stop early for success or extend weak studies hoping for a reversal.

FORMULA INTUITION

Alpha-spending methods allocate cumulative Type I error across looks, while always-valid e-values or confidence sequences use different martingale-based guarantees. The boundary depends on the chosen design.

SIMPLE MARKET EXAMPLE

Checking a weekly A/B execution test at 5% for twelve weeks creates more than 5% chance of at least one false alarm. A group-sequential boundary is stricter early and relaxes according to plan.

PYTHON / SYSTEM IMPLEMENTATION

Register look times, stopping boundaries, maximum horizon, and actions before launch. Persist every interim result, including those that did not trigger a stop.

CONFUSIONS TO AVOID

Stopping for harm and stopping for benefit may require separate boundaries. Simply reporting the p-value from the final observed week does not reveal the adaptive path.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment controller owns the clock and exposes only valid decision states: continue, stop for benefit, stop for harm, or stop for futility. Manual overrides are recorded as protocol deviations.

RELATED CONCEPTS AND QUICK RECALL

Related: alpha spending, confidence sequence, e-value, futility boundary. Recall task: Explain why weekly peeking at a fixed 5% test inflates false positives even if each calculation is correct.

Missingness, selection, and censoring

INTERMEDIATE

ONE-SENTENCE MEANING

Missing or selected observations can change the population represented by a test and bias the estimate before uncertainty is considered. Censoring hides outcome magnitude beyond a boundary and requires its own model.

MEMORY JOGGER

Ask why a row is absent, not only how many rows are absent. Missing difficult trades, failed orders, or delisted assets often makes the observed sample systematically favorable.

WHY IT MATTERS IN QUANT RESEARCH

Quant datasets lose observations through vendor outages, liquidity filters, survivorship, execution failures, and unavailable future returns. A small p-value on selected survivors cannot recover the excluded population.

FORMULA INTUITION

Complete-case inference targets $E[Y \mid \text{observed}]$ unless missingness assumptions justify broader interpretation. Inverse-probability weights use $1/P(\text{observed} \mid X)$, but rely on a credible observation model.

SIMPLE MARKET EXAMPLE

If only filled orders report slippage, the mean excludes difficult orders that never executed. Testing observed fills can overstate both quality and capacity.

PYTHON / SYSTEM IMPLEMENTATION

Publish exclusion counts and reasons by time, asset, and outcome proxy. Use missingness indicators, sensitivity bounds, and point-in-time eligibility snapshots rather than silent row drops.

CONFUSIONS TO AVOID

Imputation does not create information, and significance after aggressive filtering may describe only an easy subset. Conditioning on a variable affected by the strategy can introduce collider bias.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The data contract distinguishes unavailable, ineligible, censored, and failed outcomes. Inference artifacts include a population bridge from source frame to analyzed sample.

RELATED CONCEPTS AND QUICK RECALL

Related: missing at random, survivorship bias, censoring, inverse weighting. Recall task: Design a population bridge for an execution study where unfilled orders have no realized fill price.

Assumption checks and model criticism

INTERMEDIATE

ONE-SENTENCE MEANING

Assumption checks examine whether the test's calibration conditions are plausible enough for the intended claim. Model criticism seeks structured failures rather than using diagnostics as a ritual checklist.

MEMORY JOGGER

Every test is a model of what variation would look like under the null. Diagnostics ask where the observed data conflict with that variation mechanism.

WHY IT MATTERS IN QUANT RESEARCH

Heavy tails, autocorrelation, unequal variance, few clusters, discreteness, and nonstationarity can invalidate standard errors or reference distributions. Alternative estimators should target the actual failure mode.

FORMULA INTUITION

Residual diagnostics evaluate $E[e|X]$, variance, dependence, and influence; resampling diagnostics evaluate exchangeability or block structure. No single normality test validates the whole inferential design.

SIMPLE MARKET EXAMPLE

A mean-return t-test may show extreme influence from one crisis week and strong weekly autocorrelation. Robust scale alone does not fix dependence, while HAC alone does not neutralize a bad tick.

PYTHON / SYSTEM IMPLEMENTATION

Automate residual plots, influence contributions, lag diagnostics, cluster counts, variance ratios, and resampling stability. Define escalation actions before diagnostics are viewed.

CONFUSIONS TO AVOID

A non-significant assumption test does not prove the assumption, especially with small samples. Replacing one fragile test with many uncorrected diagnostics can also create false alarms.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The inference pipeline generates a method-specific diagnostic bundle and health state. Results with unresolved red diagnostics cannot be promoted, even when the headline threshold is crossed.

RELATED CONCEPTS AND QUICK RECALL

Related: residual analysis, exchangeability, influence, robustness check. Recall task: Map heavy tails, serial dependence, and unequal group variance to three different corrective methods.

One-sample mean test

INTERMEDIATE

ONE-SENTENCE MEANING

A one-sample mean test asks whether the population mean differs from a specified benchmark. Its validity depends on how the mean's standard error reflects tails, dependence, and the sampling design.

MEMORY JOGGER

Subtract the benchmark, average the differences, and scale by uncertainty in that average. The method tests expected level, not median outcome or probability of profit.

WHY IT MATTERS IN QUANT RESEARCH

Mean return, mean forecast error, and average implementation shortfall are common targets. Because P&L aggregates, the mean can be economically natural even when the distribution is skewed.

FORMULA INTUITION

$T = (\bar{x} - \mu_0) / (s / \sqrt{n})$ with $n-1$ degrees of freedom for IID normal observations. Large-sample use relies on a suitable central limit theorem, not on raw returns being exactly normal.

SIMPLE MARKET EXAMPLE

Twenty-four monthly net returns average 0.6% with standard deviation 1.8%. The IID standard error is 0.367%, giving t about 1.63 and substantial uncertainty despite a positive average.

PYTHON / SYSTEM IMPLEMENTATION

Compute the mean in economic units, then choose IID, HAC, cluster, or block-bootstrap uncertainty from the design. Save contribution by observation to reveal whether one period drives the result.

CONFUSIONS TO AVOID

The t -test does not test whether most periods are positive, and a large n does not cure biased selection. Annualizing the mean before dividing by an unannualized standard error corrupts units.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Backtest evaluation uses the one-sample estimate as one evidence panel beside drawdown, cost stress, and stability. The benchmark may be zero, funding, or a required hurdle and is versioned explicitly.

RELATED CONCEPTS AND QUICK RECALL

Related: sample mean, t distribution, benchmark return, contribution. Recall task: Calculate the IID t -statistic for mean 0.4%, standard deviation 2%, and 36 independent months.

Welch two-sample comparison

INTERMEDIATE

ONE-SENTENCE MEANING

Welch's test compares two population means without assuming equal variances. It uses separate variance estimates and an approximate degrees-of-freedom calculation.

MEMORY JOGGER

Compare the group means, but let each group's noise and size contribute honestly to the denominator. This is usually safer than automatically pooling variances.

WHY IT MATTERS IN QUANT RESEARCH

Researchers use two-sample contrasts for signal buckets, venues, models, or periods. The groups must still be independently sampled at the chosen unit and comparable for the target claim.

FORMULA INTUITION

$T = (\bar{x} - \bar{y}) / \sqrt{s_x^2/n_x + s_y^2/n_y}$. Welch-Satterthwaite degrees of freedom reflect the two estimated variance contributions.

SIMPLE MARKET EXAMPLE

A new venue reduces mean cost by 1.2 bp, but its cost variance is twice the old venue's and it has fewer orders. Welch uncertainty correctly gives the noisier, smaller group less precision.

PYTHON / SYSTEM IMPLEMENTATION

Use vetted library implementations and report group counts, means, variances, difference, interval, and degrees of freedom. Cluster at the assignment or sampling unit when orders share parents or dates.

CONFUSIONS TO AVOID

Welch handles unequal variances, not confounding, time drift, or dependent groups. A significant difference between periods may reflect volatility regime rather than a model change.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The comparison service requires a group-definition artifact and balance summary. Promotion relies on an adjusted contrast when observational group composition differs materially.

RELATED CONCEPTS AND QUICK RECALL

Related: heteroskedasticity, pooled variance, group contrast, confounding. Recall task: Explain why Welch's test can be preferable even when sample variances look roughly similar.

Paired and matched comparisons

INTERMEDIATE

ONE-SENTENCE MEANING

A paired test analyzes within-pair differences when two outcomes share the same observational unit or matched context. Pairing removes common variation and targets the average difference directly.

MEMORY JOGGER

Compare like with like before averaging. Good pairing can reveal a small treatment effect that is hidden by large cross-unit variation.

WHY IT MATTERS IN QUANT RESEARCH

Execution algorithms can be replayed on the same orders, forecasts compared on the same dates, and portfolio variants evaluated under matched scenarios. Pairing is powerful only when pairs are legitimate and timing is symmetric.

FORMULA INTUITION

Define $d_i = x_i - y_i$ and test $H_0: E[d] = 0$ using $T = \bar{d} / (s_d / \sqrt{n_{\text{pairs}}})$. The relevant sample size is complete independent pairs, not the total number of cells.

SIMPLE MARKET EXAMPLE

Two models have similar average forecast loss across different months, but on each shared month model B is consistently lower. A paired analysis captures the within-month improvement.

PYTHON / SYSTEM IMPLEMENTATION

Build pairs with stable keys and report unmatched counts and reasons. Use clustered or block uncertainty if pair differences remain dependent across assets or adjacent dates.

CONFUSIONS TO AVOID

Dropping incomplete pairs can create selection bias, and pairing unrelated observations only to reduce variance is invalid. A paired t-test still targets mean difference and remains sensitive to extreme differences.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment schema declares pair key, ordering, match tolerance, and exclusion policy. Reconciliation verifies that both variants consumed identical inputs before comparing outcomes.

RELATED CONCEPTS AND QUICK RECALL

Related: difference score, matched design, blocking, replay test. Recall task: Design the pair key for comparing two transaction-cost models on historical parent orders.

Proportion and hit-rate tests

INTERMEDIATE

ONE-SENTENCE MEANING

A proportion test evaluates the probability of a binary outcome such as a win, breach, or correct sign. It summarizes frequency, not payoff magnitude or economic value.

MEMORY JOGGER

Convert each eligible unit into success or failure under a fixed rule, then estimate uncertainty in the success rate. The rule and denominator are part of the parameter.

WHY IT MATTERS IN QUANT RESEARCH

Hit rate, fill probability, stop frequency, and directional accuracy are interpretable operational metrics. They should be paired with conditional payoff because many small wins can coexist with rare large losses.

FORMULA INTUITION

For independent Bernoulli outcomes, $SE(\hat{p}) = \sqrt{p_0(1-p_0)/n}$ under the null. Exact binomial, score, or Wilson methods are preferable to fragile normal approximations near boundaries or small n .

SIMPLE MARKET EXAMPLE

A strategy wins on 58 of 100 trades, but its losers average four times the winners. The hit-rate evidence may be real while expected P&L remains negative.

PYTHON / SYSTEM IMPLEMENTATION

Define success before observing results, retain counts rather than rounded percentages, and use cluster-aware methods for repeated signals. Report Wilson intervals and payoff summaries together.

CONFUSIONS TO AVOID

Testing overlapping daily directional labels as independent inflates precision. A 50% benchmark may also be inappropriate when class imbalance or abstention changes the denominator.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The metric registry stores event definition, eligible population, abstention handling, and cluster unit. Strategy review blocks any hit-rate claim that omits average gain, average loss, and tail exposure.

RELATED CONCEPTS AND QUICK RECALL

Related: Bernoulli outcome, Wilson interval, class balance, payoff ratio. Recall task: Explain why a statistically significant 55% hit rate can still describe a losing strategy.

Permutation and randomization tests

INTERMEDIATE

ONE-SENTENCE MEANING

A permutation test compares the observed statistic with values generated by rearrangements that would be exchangeable under the null. It can provide finite-sample calibration when the randomization or symmetry is credible.

MEMORY JOGGER

Break the relationship in exactly the way the null says should not matter. Preserve every structure that would still exist under that null.

WHY IT MATTERS IN QUANT RESEARCH

Permutation methods are useful for unusual metrics and small samples where parametric reference distributions are doubtful. Market time dependence means naive row shuffling is rarely appropriate.

FORMULA INTUITION

With B valid rearrangements, $p_{\text{hat}} = (1 + \text{number}[T_b \text{ at least as extreme as } T_{\text{obs}}]) / (B + 1)$. The plus-one correction avoids zero p-values from finite simulation.

SIMPLE MARKET EXAMPLE

To test a cross-sectional signal, shuffle ranks within each date rather than across all asset-days. This preserves date shocks and universe size while severing the same-day signal-return link.

PYTHON / SYSTEM IMPLEMENTATION

Write a null generator that operates on blocks, dates, pairs, or assignment labels as designed. Seed reproducibly, store permutation summaries, and validate that protected structures remain unchanged.

CONFUSIONS TO AVOID

Permutation does not remove bias from a selected statistic or leaked features. Arbitrary shuffling can destroy autocorrelation, seasonality, and clusters, producing an unrealistically narrow null.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The resampling service versions the exchangeability rule and runs invariant checks before accepting results. Method reviewers approve the null generator separately from the statistic code.

RELATED CONCEPTS AND QUICK RECALL

Related: exchangeability, randomization inference, block shuffle, Monte Carlo p-value. Recall task: Design a valid permutation scheme for a daily cross-sectional factor with shared market shocks.

Bootstrap inference

INTERMEDIATE

ONE-SENTENCE MEANING

The bootstrap approximates an estimator's sampling distribution by repeatedly resampling observed information units. Its validity depends on resampling in a way that reproduces the dependence and heterogeneity relevant to the estimator.

MEMORY JOGGER

Treat the sample as a stand-in population, draw many new samples, and watch the estimate vary. Resample clusters or blocks when rows are not independent.

WHY IT MATTERS IN QUANT RESEARCH

Bootstrap intervals support nonlinear metrics such as Sharpe, drawdown, turnover-adjusted return, and rank spread. They are especially useful when analytic standard errors are cumbersome.

FORMULA INTUITION

For bootstrap estimates θ_{hat_b} , uncertainty comes from their empirical distribution. Percentile, basic, studentized, and BCa intervals make different accuracy assumptions and can disagree.

SIMPLE MARKET EXAMPLE

Resampling individual days for a trend strategy breaks persistence and understates path risk. Moving-block resampling preserves local sequences and produces more realistic performance variation.

PYTHON / SYSTEM IMPLEMENTATION

Choose resampling unit and block length from the design, not from the desired interval. Save seeds, B, failed replicates, endpoint method, and sensitivity across plausible block lengths.

CONFUSIONS TO AVOID

The bootstrap cannot reconstruct unobserved crises or fix a biased sample. Very small datasets yield a discrete, overly optimistic empirical population no matter how many replicates are drawn.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Validation jobs generate bootstrap distributions as stored artifacts used for uncertainty, stress views, and decision probabilities. Production monitoring compares live estimates with the same resampling contract.

RELATED CONCEPTS AND QUICK RECALL

Related: block bootstrap, BCa interval, path dependence, resampling unit. Recall task: Explain why increasing bootstrap replicates reduces simulation noise but not the sample's fundamental uncertainty.

Sign, rank, and robust tests

INTERMEDIATE

ONE-SENTENCE MEANING

Sign and rank procedures use direction or order information rather than raw magnitudes. They reduce sensitivity to extremes but target parameters that differ from the mean.

MEMORY JOGGER

When magnitudes are unreliable, ask whether observations tend to lie above a benchmark or whether one group tends to rank higher. Robustness comes from discarding some metric information.

WHY IT MATTERS IN QUANT RESEARCH

Heavy-tailed returns, latency, and slippage can make rank-based evidence more stable. Quant decisions still need raw economic magnitude because a consistent tiny improvement may not pay costs.

FORMULA INTUITION

The sign test uses the binomial count above the benchmark and targets median-like probability. Wilcoxon signed-rank adds magnitude ranks under symmetry assumptions about paired differences.

SIMPLE MARKET EXAMPLE

Nine of twelve matched months favor model B, but one month has a very large loss. The sign test highlights directional consistency while the mean difference exposes economic tail damage.

PYTHON / SYSTEM IMPLEMENTATION

Handle ties explicitly, report the estimand, and pair rank results with raw-effect intervals. Use cluster-aware resampling when signs or ranks share common dates.

CONFUSIONS TO AVOID

Nonparametric does not mean assumption-free, and rank tests do not test equality of means in general. Transforming to ranks can hide capacity-relevant magnitude differences.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence report labels robust challengers by target parameter and compares their conclusion with the primary mean-based analysis. Disagreement triggers influence and tail review.

RELATED CONCEPTS AND QUICK RECALL

Related: median, Mann-Whitney, Wilcoxon, influence resistance. Recall task: Interpret a positive sign-test result alongside a negative mean difference caused by one severe month.

Regression coefficient tests

INTERMEDIATE

ONE-SENTENCE MEANING

A regression coefficient test evaluates whether a conditional linear association differs from a benchmark within a specified model. Its meaning depends on included controls, functional form, and the source of identifying variation.

MEMORY JOGGER

A coefficient is a slope holding modeled controls fixed. The test inherits every modeling choice that defines that conditional comparison.

WHY IT MATTERS IN QUANT RESEARCH

Factor returns, predictive slopes, abnormal performance, and treatment interactions are frequently tested through regression. Standard errors must match serial, cross-sectional, and cluster structure.

FORMULA INTUITION

For coefficient β_j , $T = (\hat{\beta}_j - \beta_0) / \text{SE}(\hat{\beta}_j)$. Ordinary least squares formulas assume correctly specified conditional mean and a particular error structure unless robust methods replace them.

SIMPLE MARKET EXAMPLE

A signal coefficient falls after controlling for size and liquidity, suggesting the raw association partly reflected those exposures. The adjusted coefficient is not automatically causal if omitted variables remain.

PYTHON / SYSTEM IMPLEMENTATION

Version the design matrix, transformations, reference categories, weights, and covariance estimator. Save residual and leverage diagnostics plus coefficient stability across defensible specifications.

CONFUSIONS TO AVOID

A significant intercept is not necessarily skill when factors, costs, or timing are misspecified. Adding controls after seeing the result changes the hypothesis family and can induce multicollinearity.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The model registry links each coefficient claim to feature lineage and a frozen formula. Scoring code reproduces the same transformations and category levels used in inference.

RELATED CONCEPTS AND QUICK RECALL

Related: OLS, alpha, conditional association, omitted-variable bias. Recall task: Explain how a significant univariate factor slope can disappear after adding liquidity and sector controls.

HAC and cluster-robust uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

Heteroskedasticity-and-autocorrelation-consistent and cluster-robust estimators adjust coefficient uncertainty for structured residual dependence. They leave point estimates unchanged while changing how much independent evidence the sample contains.

MEMORY JOGGER

Keep the estimate, repair the covariance model. Choose HAC for ordered time dependence and clustering for shared shocks within known groups.

WHY IT MATTERS IN QUANT RESEARCH

Return regressions often have serially correlated, changing volatility; panels share asset and date effects. Naive errors can turn ordinary noise into apparently strong factors.

FORMULA INTUITION

HAC covariance sums weighted residual autocovariances through lag L . Cluster covariance aggregates score contributions within clusters, allowing arbitrary dependence inside each cluster under large-cluster asymptotics.

SIMPLE MARKET EXAMPLE

A daily alpha t-stat falls from 3.1 to 1.8 after using a 20-day HAC lag. The return estimate is identical, but overlapping positions had overstated precision.

PYTHON / SYSTEM IMPLEMENTATION

Declare lag or cluster policy before results, report number of clusters, and use small-sample corrections where supported. Compare alternative reasonable lags and cluster levels as sensitivity analysis.

CONFUSIONS TO AVOID

Robust standard errors do not fix biased coefficients, leakage, or too few clusters. Extremely long HAC lags can add noise, while short lags can miss the dependence they were meant to address.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Inference policy maps each data topology to approved covariance estimators and minimum cluster counts. The pipeline fails closed when the design lacks enough independent groups.

RELATED CONCEPTS AND QUICK RECALL

Related: Newey-West, sandwich covariance, two-way clustering, residual dependence. Recall task: Choose between HAC, asset clustering, date clustering, and two-way clustering for a panel factor regression.

Confidence intervals and test duality

INTERMEDIATE

ONE-SENTENCE MEANING

A confidence interval gives the range of parameter values not rejected by a corresponding family of tests at the stated level. It communicates magnitude and precision more directly than a thresholded p-value.

MEMORY JOGGER

Look at the entire plausible range, not only whether it crosses zero. The interval can reveal that both valuable gains and unacceptable losses remain consistent with the data.

WHY IT MATTERS IN QUANT RESEARCH

Intervals help separate absence of evidence from evidence of a negligible effect. They also expose when a significant estimate is too small to matter economically.

FORMULA INTUITION

A two-sided $1-\alpha$ interval often has form $\hat{\theta} \pm \text{critical} \times \text{SE}$. If the interval excludes θ_0 , the matching two-sided test rejects θ_0 at level α .

SIMPLE MARKET EXAMPLE

An alpha estimate is 18 bp monthly with 95% interval [2,34]. Zero is excluded, yet the interval crosses a 15 bp deployment hurdle and leaves economic value uncertain.

PYTHON / SYSTEM IMPLEMENTATION

Compute intervals with the same dependence, multiplicity, and resampling assumptions as the test. Display raw units, benchmark, and practical threshold on the same visual scale.

CONFUSIONS TO AVOID

A 95% interval does not assign 95% probability to a fixed parameter under frequentist interpretation. Selecting only intervals from significant studies destroys their nominal coverage.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Decision dashboards use intervals as the primary evidence object and p-values as secondary calibration. Gates inspect lower bounds against economic thresholds rather than merely checking exclusion of zero.

RELATED CONCEPTS AND QUICK RECALL

Related: coverage, uncertainty band, critical value, practical threshold. Recall task: Interpret an interval that excludes zero but includes both sub-hurdle and profitable effects.

Backtest significance and path dependence

INTERMEDIATE

ONE-SENTENCE MEANING

Backtest significance asks whether observed strategy performance exceeds a defensible null after accounting for search, dependence, costs, and path construction. A backtest is a selected time path, not a simple random sample of trades.

MEMORY JOGGER

Test the complete research process, not only the final return vector. The null should reproduce signal search, overlapping holdings, rebalance rules, and portfolio constraints that create apparent performance.

WHY IT MATTERS IN QUANT RESEARCH

Standard t-tests can understate uncertainty when returns overlap or when the strategy was chosen from many variants. Drawdown and capacity also depend on order, which a mean-only test ignores.

FORMULA INTUITION

One statistic may use mean net return divided by HAC uncertainty; a stronger null reruns the selection process on block-resampled or signal-randomized data. The maximum selected performance is the relevant benchmark.

SIMPLE MARKET EXAMPLE

A strategy with $t=2.4$ was the best of 500 parameter sets. Under a search-aware null, comparable maxima occur often, so the isolated t-stat overstates evidence.

PYTHON / SYSTEM IMPLEMENTATION

Log the complete variant grid, net-return construction, and selection rule. Recompute the entire pipeline within each valid resample and reserve a sealed forward period for confirmation.

CONFUSIONS TO AVOID

Trade count is not independent sample size, and gross significance is not net significance. Reordering returns during resampling can erase drawdown and turnover interactions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The backtest engine emits an evidence bundle containing search-adjusted null, temporal validation, cost stress, and path diagnostics. Promotion requires consistency across all components.

RELATED CONCEPTS AND QUICK RECALL

Related: data snooping, reality check, block resampling, walk-forward validation. Recall task: Explain why testing only the winner of a 500-model search against a single-model null is miscalibrated.

Sharpe ratio uncertainty

INTERMEDIATE

ONE-SENTENCE MEANING

Sharpe ratio uncertainty reflects estimation error in both average excess return and volatility. Serial correlation, nonnormality, short histories, and selection can make a high point estimate much less convincing than it appears.

MEMORY JOGGER

Sharpe is a ratio of two noisy quantities. Treat it as an estimate with a distribution, not as a permanent property printed to two decimals.

WHY IT MATTERS IN QUANT RESEARCH

Capital allocation and model comparison often overreact to small Sharpe differences. Intervals and probability of exceeding a required Sharpe provide a more honest view of ranking uncertainty.

FORMULA INTUITION

$SR_hat = \text{mean}(r-f) / \text{sd}(r-f)$, with annualization depending on the return process. IID normal approximations exist, but HAC, block bootstrap, or specialized corrections are often more credible.

SIMPLE MARKET EXAMPLE

Two strategies have Sharpe 1.1 and 0.9 over three years, but their wide overlapping intervals make the ranking unresolved. Costs or one crisis month can reverse it.

PYTHON / SYSTEM IMPLEMENTATION

Compute Sharpe from a sealed net return series and bootstrap complete paths with blocks. Report annualization convention, interval, downside metrics, and leave-one-period sensitivity.

CONFUSIONS TO AVOID

A significant mean does not imply a precisely estimated Sharpe, and annualized daily Sharpe can be inflated by autocorrelation. Selecting the highest historical Sharpe adds winner's curse.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Portfolio review consumes a distribution over plausible Sharpe and applies conservative allocation haircuts. Live monitoring updates evidence without overwriting the original research estimate.

RELATED CONCEPTS AND QUICK RECALL

Related: information ratio, annualization, ratio estimator, probabilistic Sharpe. Recall task: Explain why Sharpe rankings based on three years of monthly data can be unstable even when point estimates differ.

Cross-sectional factor evidence

INTERMEDIATE

ONE-SENTENCE MEANING

Cross-sectional factor evidence tests whether asset characteristics predict relative future outcomes across a point-in-time universe. Inference must handle common date shocks, repeated assets, sector structure, and portfolio formation choices.

MEMORY JOGGER

The factor observation is not one stock-day in isolation. Information arrives through repeated cross-sections whose membership and common shocks must remain visible.

WHY IT MATTERS IN QUANT RESEARCH

Factor research can produce huge row counts and deceptively small naive errors. Breadth, turnover, neutralization, and weighting affect both the estimand and deployable return.

FORMULA INTUITION

A Fama-MacBeth approach estimates cross-sectional slopes each date and tests their time-series mean. Portfolio-spread approaches test date-level long-short returns with time-dependent uncertainty.

SIMPLE MARKET EXAMPLE

A value slope is positive across most months but concentrated in microcaps. Liquidity weighting and sector-neutral construction may reduce both magnitude and significance.

PYTHON / SYSTEM IMPLEMENTATION

Build point-in-time universes, lag features, preserve delistings, and store date-level coefficients or spreads. Use date-clustered, two-way, or time-series inference appropriate to the estimator.

CONFUSIONS TO AVOID

Treating all asset-days as IID is invalid, and neutralizing after inspecting performance adds hidden search. Survivorship and stale fundamentals can create significance before the statistical test begins.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The factor registry links definition, universe, neutralization, weighting, horizon, and inference method. Live factor monitoring decomposes decay into breadth, magnitude, turnover, and universe mix.

RELATED CONCEPTS AND QUICK RECALL

Related: Fama-MacBeth, rank IC, long-short spread, point-in-time universe. Recall task: Describe two valid inference routes for a cross-sectional factor and the independence unit each uses.

Event studies and abnormal returns

INTERMEDIATE

ONE-SENTENCE MEANING

An event study measures returns around identified events relative to an expected-return model. Valid inference requires careful event timing, overlap control, benchmark estimation, and treatment of clustered news.

MEMORY JOGGER

Define what return would have occurred without the event, then compare the observed path around a fixed timestamp. The benchmark model is part of the hypothesis.

WHY IT MATTERS IN QUANT RESEARCH

Earnings, rebalances, policy releases, and corporate actions can reveal information response and execution risk. Event clustering and leakage before timestamps often dominate naive significance.

FORMULA INTUITION

Abnormal return $AR_{it} = R_{it} - E[R_{it} | \text{model}]$, and cumulative abnormal return sums AR over a declared window. Cross-event tests must reflect correlation among firms and calendar dates.

SIMPLE MARKET EXAMPLE

Several banks announce related news on the same day, producing correlated abnormal returns. Counting each bank as independent exaggerates evidence for an industry-wide event effect.

PYTHON / SYSTEM IMPLEMENTATION

Store source timestamp, timezone, announcement phase, estimation window, event window, benchmark version, and overlap flags. Cluster by event date or use randomization aligned with the event assignment mechanism.

CONFUSIONS TO AVOID

Moving the event window after viewing a price chart creates multiplicity. Overnight announcements cannot be aligned mechanically to the prior close without market-session logic.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event service normalizes timestamps and emits a canonical panel with benchmark residuals and clustering keys. Strategy research consumes that panel rather than rebuilding windows ad hoc.

RELATED CONCEPTS AND QUICK RECALL

Related: cumulative abnormal return, benchmark model, event clustering, leakage. Recall task: Design an event window and cluster rule for earnings announcements released before open and after close.

Subgroups, interactions, and regime claims

INTERMEDIATE

ONE-SENTENCE MEANING

A subgroup claim concerns how an effect changes across pre-specified states, while an interaction test evaluates the difference between subgroup effects directly. Separate significance within one group and not another does not prove a difference.

MEMORY JOGGER

Test the gap between effects, not the gap between p-values. Regime conclusions need enough observations and labels available at decision time.

WHY IT MATTERS IN QUANT RESEARCH

Signals often behave differently across volatility, liquidity, or trend states. Conditional deployment can help, but searching many states creates multiplicity and unstable boundaries.

FORMULA INTUITION

In $Y = \alpha + \beta X + \gamma S + \delta(X \text{ times } S) + e$, δ tests effect modification by state S . The subgroup effects are β and $\beta + \delta$ with correlated uncertainty.

SIMPLE MARKET EXAMPLE

A signal is significant in calm markets and insignificant in stress, yet the interaction interval includes zero. The apparent contrast may be sampling noise from a small stress sample.

PYTHON / SYSTEM IMPLEMENTATION

Register state definitions, cutoffs, and interaction contrasts before evaluation. Report state counts, transition behavior, and pooled-shrinkage estimates when segments are thin.

CONFUSIONS TO AVOID

Data-mined regimes can explain any history, and separate within-group tests are not an interaction test. Conditioning on a future-defined regime leaks information into deployment.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The regime controller uses only approved point-in-time labels and uncertainty-aware conditional effects. It defaults to conservative exposure when the live state is ambiguous or unsupported.

RELATED CONCEPTS AND QUICK RECALL

Related: effect modification, interaction, subgroup analysis, regime classifier. Recall task: Explain why $p < 0.05$ in calm markets and $p > 0.05$ in stress does not establish a regime difference.

Equivalence and noninferiority

INTERMEDIATE

ONE-SENTENCE MEANING

Equivalence tests ask whether an effect lies within a predeclared practically negligible range, while noninferiority tests ask whether a new method is not worse than a margin. They are designed to support bounded similarity claims that ordinary tests cannot.

MEMORY JOGGER

Failure to find a difference is not evidence of sameness. To claim similarity, show the uncertainty range fits inside an acceptable margin.

WHY IT MATTERS IN QUANT RESEARCH

Faster models, cheaper data, and simpler execution may be valuable if performance loss is bounded. Equivalence and noninferiority connect statistical evidence directly to an operational tradeoff.

FORMULA INTUITION

Equivalence commonly uses two one-sided tests against $-\Delta$ and $+\Delta$. Noninferiority tests $H_0: \theta \leq -\Delta$ versus $H_1: \theta > -\Delta$ under a clear sign convention.

SIMPLE MARKET EXAMPLE

A new risk model is 30% faster, and its tracking-error difference interval is $[-3, +4]$ bp against an equivalence margin of ± 5 bp. The data support practical equivalence at that margin.

PYTHON / SYSTEM IMPLEMENTATION

Set the margin from business tolerance before results and justify it in raw units. Use both intention-to-deploy and protocol-compliant analyses when operational deviations matter.

CONFUSIONS TO AVOID

Choosing a wide margin after seeing data makes the claim meaningless. An underpowered superiority test that fails to reject zero does not establish equivalence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The replacement-model gate requires an approved margin, interval method, latency or cost benefit, and rollback plan. Deployment starts with monitoring against the same margin.

RELATED CONCEPTS AND QUICK RECALL

Related: TOST, practical margin, superiority, replacement model. Recall task: Define a noninferiority margin for a cheaper market-data feed and state which failure would block migration.

Bayesian evidence and frequentist tests

INTERMEDIATE

ONE-SENTENCE MEANING

Bayesian inference updates a probability distribution over parameters using a prior and likelihood, while frequentist tests calibrate procedures over repeated samples under specified nulls. They answer related but different questions.

MEMORY JOGGER

Frequentist evidence asks how surprising data are under a reference; Bayesian evidence asks how beliefs change under a model. Neither framework removes the need for valid data and design.

WHY IT MATTERS IN QUANT RESEARCH

Posterior probabilities and shrinkage can be useful for portfolio decisions, especially across many weak signals. Frequentist error control remains valuable for governed discovery pipelines and repeated experimentation.

FORMULA INTUITION

Posterior $p(\theta|\text{data})$ is proportional to likelihood times prior. A Bayes factor compares marginal likelihoods, while a p-value is a tail area and cannot be converted into a posterior null probability without additional assumptions.

SIMPLE MARKET EXAMPLE

A noisy alpha estimate may have $p=0.04$ but only modest posterior probability of exceeding the 20 bp hurdle under a skeptical prior. The conclusions differ because the questions differ.

PYTHON / SYSTEM IMPLEMENTATION

Version priors, likelihood, and posterior diagnostics just as carefully as test specifications. When reporting both frameworks, align estimands and show why the prior reflects a defensible signal base rate.

CONFUSIONS TO AVOID

A default prior is still a modeling choice, and a Bayes factor is sensitive to prior scale. Frequentist significance and posterior probability should not be mixed into one unlabeled score.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The evidence layer can store test calibration and decision posterior side by side. Governance defines which object supports discovery control and which supports capital sizing.

RELATED CONCEPTS AND QUICK RECALL

Related: prior, likelihood, posterior, Bayes factor. Recall task: Explain why $p=0.03$ does not imply a 97% posterior probability that alpha is positive.

Evidence registry and production gates

INTERMEDIATE

ONE-SENTENCE MEANING

An evidence registry records the complete lineage from hypothesis through data, estimator, uncertainty, diagnostics, and decision. Production gates translate that artifact into repeatable actions without reducing evidence to one threshold.

MEMORY JOGGER

If a result cannot be replayed, it cannot be governed. If a gate cannot explain why it passed, it cannot safely control capital.

WHY IT MATTERS IN QUANT RESEARCH

Quant systems need continuity between research, validation, deployment, and monitoring. Registry metadata prevents a live metric from drifting away from the definition that earned approval.

FORMULA INTUITION

A gate is a conjunction such as lower confidence bound > economic hurdle, adjusted evidence passes, stability diagnostics green, and cost stress acceptable. Each component has a version and owner.

SIMPLE MARKET EXAMPLE

A signal passes raw significance but fails after-cost lower-bound and regime-stability checks. The registry preserves the promising research result while correctly blocking capital deployment.

PYTHON / SYSTEM IMPLEMENTATION

Store immutable artifact IDs for data snapshot, code commit, specification, search family, estimates, intervals, diagnostics, reviewer decision, and expiry. Build replay and diff tools around those IDs.

CONFUSIONS TO AVOID

A green p-value cannot override failed lineage, stale data, or unresolved leakage. Production monitoring must not silently refit thresholds using the same incidents it evaluates.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane evaluates approved gates, assigns warning or degraded states, and supports rollback independently of model releases. New evidence creates a new artifact rather than editing history.

RELATED CONCEPTS AND QUICK RECALL

Related: model registry, lineage, promotion policy, reproducibility. Recall task: Design a minimum gate for promoting a statistically promising but high-turnover signal to paper trading.

From idea to governed evidence

INTERMEDIATE

DESIGN PRINCIPLE

The statistical result is one stage in an evidence supply chain. Every transition can create leakage, selection, unit mismatch, or decision drift if its contract is implicit.

1

Research idea

Mechanism, target action, expected sign, failure conditions

2

Registered study

Estimand, hypotheses, useful effect, family, stopping rule

3

Analysis snapshot

Point-in-time data, eligibility, grain, outcomes, clusters

4

Estimate and uncertainty

Magnitude, interval, statistic, adjusted evidence, diagnostics

5

Robustness challenge

Holdout, placebo, alternative estimator, cost and regime stress

6

Decision gate

Reject, continue research, paper trade, limited pilot, or promote

7

Production evidence

Live definition parity, calibration, expiry, rollback, postmortem

AUDIT CHECKPOINT

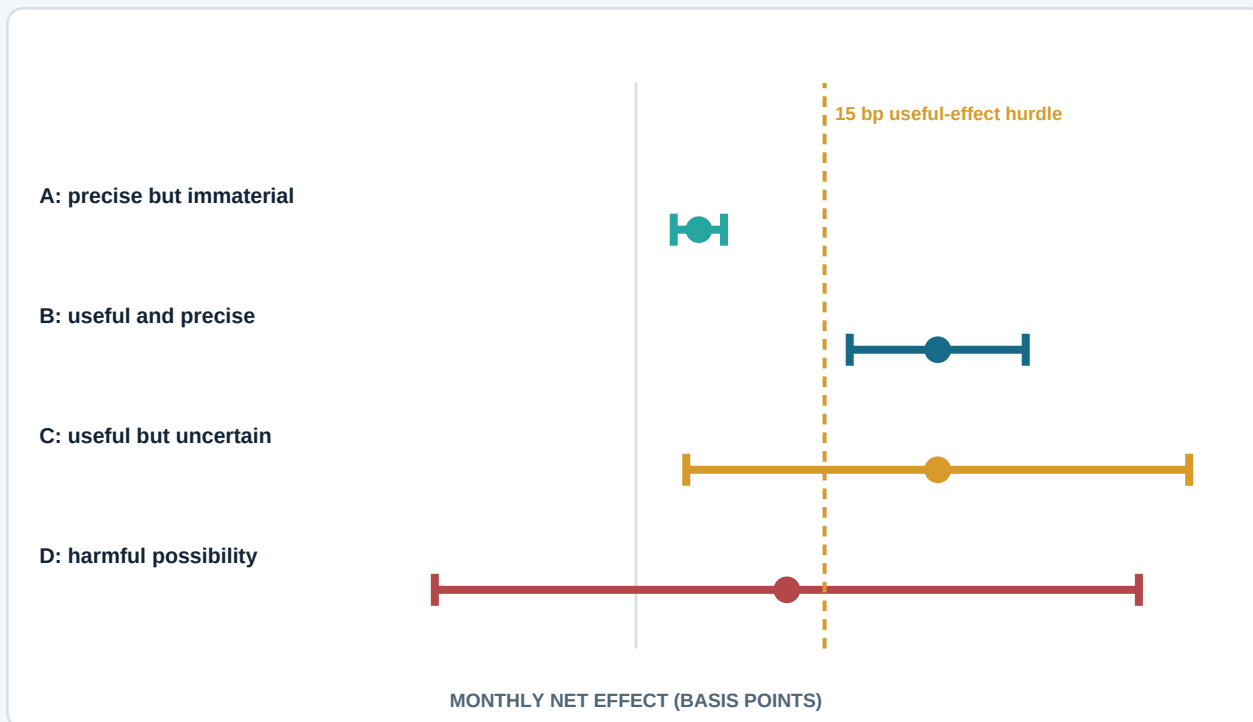
Replay one promoted result from immutable inputs to final gate. Verify that displayed hypothesis, estimator, correction, cost model, and production metric all share the same artifact IDs.

Effect size, uncertainty, and p-value

INTERMEDIATE

READING RULE

The same estimated effect can produce different p-values when uncertainty changes, and the same p-value can accompany effects with very different economic value.



INTERPRETATION

A may reject zero yet fail the economic hurdle. B supports both statistical and practical claims. C may be promising but needs more information. D leaves a broad range including harm, so threshold language alone would conceal decision risk.

REPORT DESIGN

Plot the estimate, interval, null, and practical threshold on one scale. Add sample span, dependence method, and multiplicity status in the caption so precision cannot be detached from its source.

RECALL

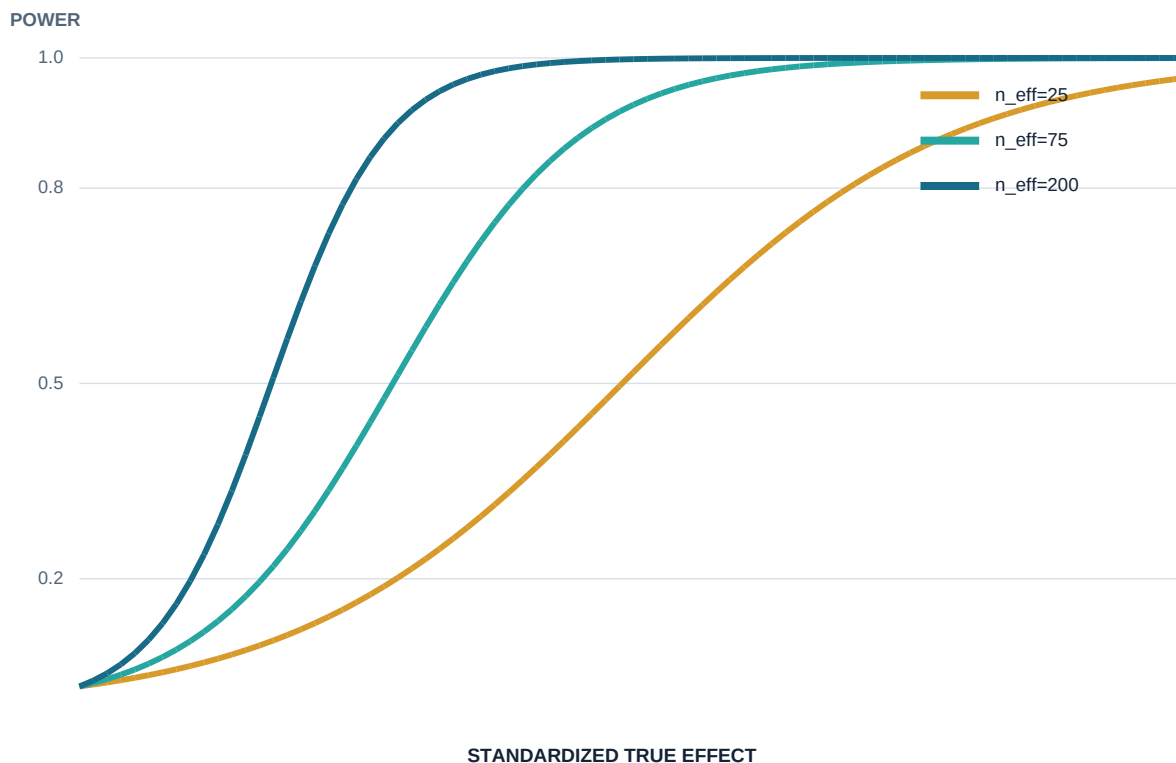
A p-value answers a tail-area question under the null; an interval answers which parameter values remain compatible with the procedure; the hurdle answers whether the magnitude changes action.

Power, effect size, and information

INTERMEDIATE

SYNTHETIC DESIGN

The curves show detection probability as true standardized effect grows. More independent observations shift the curve left; redundant rows do not provide the same gain.



PLANNING INTERPRETATION

Select a minimum useful effect on the horizontal axis, then ask whether expected independent information reaches the required detection probability. If not, redesign measurement, pairing, or horizon before collecting more redundant rows.

DEPENDENCE PENALTY

Autocorrelation, clustering, concentration, and overlapping labels lower effective information. Power simulations should generate those structures explicitly instead of replacing raw n with an optimistic scalar.

DECISION USE

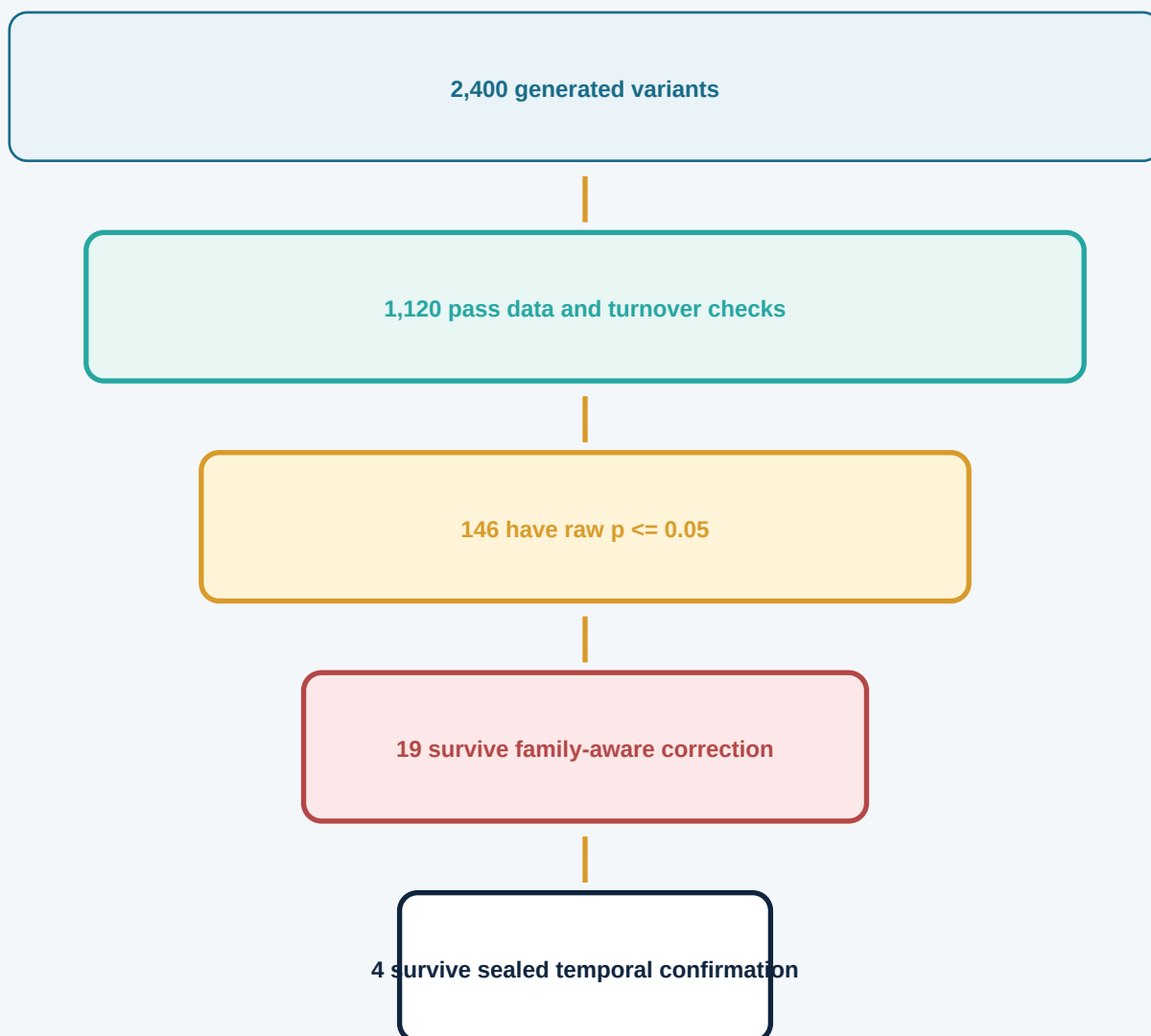
A study unable to detect the smallest deployable effect cannot support a negative conclusion. Label it inconclusive and specify the additional independent time, cross-sectional breadth, or variance reduction needed.

From signal search to false discoveries

INTERMEDIATE

SCENARIO

A library evaluates many correlated feature, horizon, universe, and neutralization variants. The final reported p-value must be interpreted through that selection path.



WHAT CORRECTION CANNOT DO

Multiplicity control calibrates a defined family, but it does not fix leakage, implausible costs, unstable data, or a family narrowed after seeing results. Independent confirmation remains valuable because it tests the selected pipeline as one frozen claim.

REQUIRED ARTIFACT

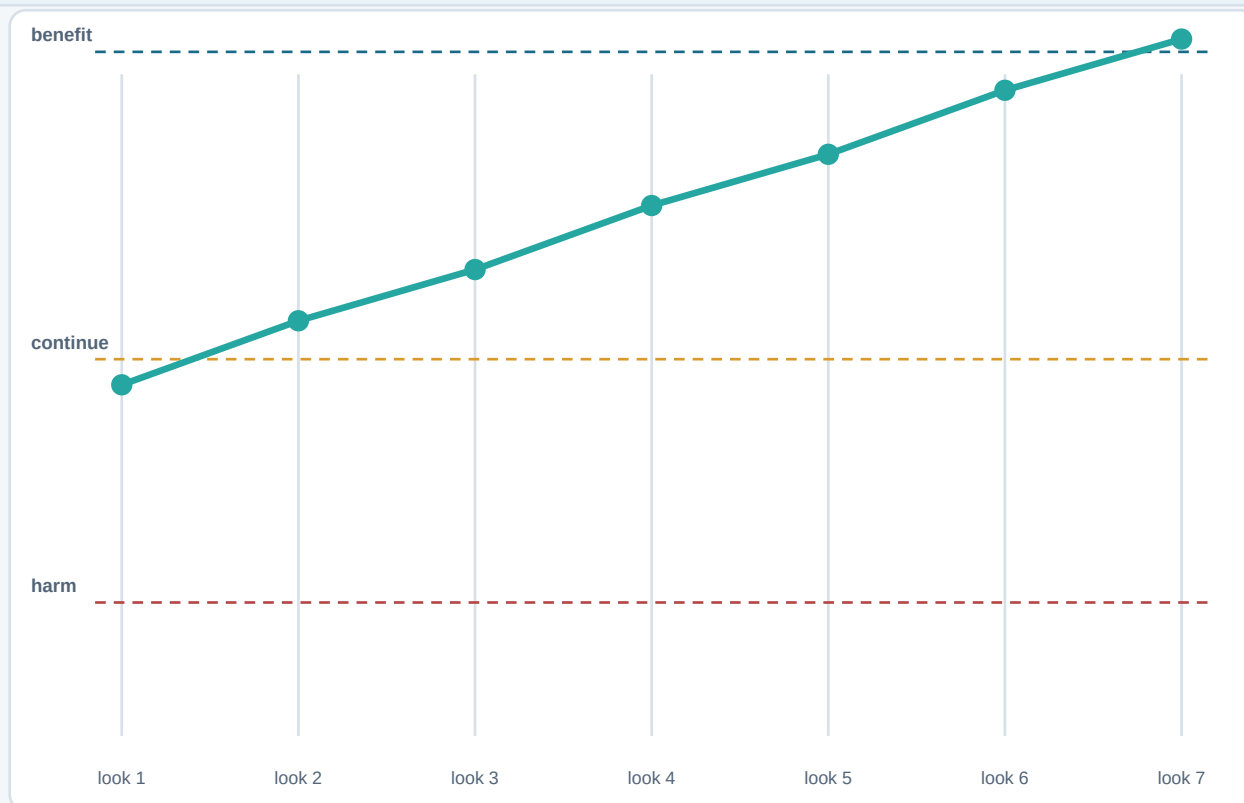
Keep the search ledger, family ID, raw and adjusted evidence, selection rule, discarded candidates, and confirmation result. A result without its unsuccessful siblings is not reviewable evidence.

Sequential evidence and stopping rules

INTERMEDIATE

MONITORING DESIGN

The pilot is reviewed at predeclared information times. Early success requires stronger evidence, while harm and futility boundaries protect capital and time.



INTERPRETATION

The evidence path approaches but does not cross the early benefit boundary until the final planned look. A fixed 5% test repeated at every point would allow an earlier false-positive-prone stop.

OPERATIONAL CONTRACT

The experiment controller records each look, current information, valid boundary, and permitted action. Missing a scheduled look or changing maximum duration creates a protocol deviation requiring recalibration.

SAFETY

Benefit, harm, and futility answer different questions and can use asymmetric thresholds. A pilot can stop for material loss even when superiority evidence remains inconclusive.

Choosing an inference method

INTERMEDIATE

START FROM TOPOLOGY

Choose the method from estimand, sampling unit, dependence, and null mechanism. Do not select the test whose assumptions are most convenient after viewing results.

QUESTION	PRIMARY METHOD	KEY CONDITION	PAIR WITH
One mean vs hurdle	Mean test	IID, HAC, block	Tail influence
Two unpaired means	Welch contrast	Independent groups	Composition bias
Matched outcomes	Paired difference	Valid complete pairs	Pair selection
Binary frequency	Binomial or score	Bernoulli clusters	Payoff magnitude
Unusual statistic	Permutation	Valid exchangeability	Broken time structure
Path metric	Block bootstrap	Representative blocks	Unseen crises
Conditional slope	Regression test	Specified mean model	Omitted variables
Many candidates	FDR plus holdout	Defined full family	Hidden search
Similarity claim	Equivalence / NI	Defensible margin	Underpowered null

ESCALATION RULE

If analytic, robust, and design-based methods disagree materially, investigate support, influence, timing, and null construction. Method disagreement is evidence about model sensitivity, not a reason to average p-values.

Formula sheet and interpretation map

INTERMEDIATE

STANDARDIZED STATISTIC

$T = (\hat{\theta} - \theta_0) / \text{SE}(\hat{\theta})$. The denominator must represent the actual independent information in the study.

ONE-SAMPLE MEAN

$T = (\bar{x} - \mu_0) / (s / \sqrt{n})$ under IID assumptions. Replace uncertainty, not the estimand, when dependence requires HAC or blocks.

WELCH CONTRAST

$T = (\bar{x} - \bar{y}) / \sqrt{s_x^2/n_x + s_y^2/n_y}$. Separate variance contributions avoid an unnecessary equal-variance assumption.

POWER SCALE

Signal-to-noise grows roughly as $\delta \sqrt{n_{\text{eff}}} / \sigma$. The target effect and effective information, not raw row count, drive detection.

FAMILY-WISE CONTROL

Bonferroni uses α/m for m claims. It controls any false rejection in the declared family under broad dependence conditions.

MONTE CARLO P-VALUE

$p_{\text{hat}} = (1 + \text{extreme resamples}) / (B + 1)$. The null generator and exchangeability rule determine validity more than the replicate count.

CONFIDENCE INTERVAL

$\hat{\theta} \pm \text{critical times SE}$ is a common symmetric form. Resampling and robust methods may produce asymmetric or differently calibrated endpoints.

EQUIVALENCE

Reject $\theta \leq -\Delta$ and $\theta \geq +\Delta$ with two one-sided tests. Both rejections are needed to support the chosen equivalence margin.

Worked calculation: mean return evidence

INTERMEDIATE

1. REGISTERED CLAIM

A monthly long-short strategy must exceed 15 bp net to justify operational cost. $H_0: \mu=0$ is the scientific null; the practical hurdle is +0.15%.

2. OBSERVED ESTIMATE

Across 48 months, mean net return is 0.42% and sample standard deviation is 1.80%. Assume independence only for this arithmetic illustration.

3. STANDARD ERROR

$SE = 1.80\% / \sqrt{48} = 0.260\%$. The estimate contains only 1.62 standard errors above zero, so its apparent magnitude is uncertain.

4. TEST STATISTIC

$t = 0.42 / 0.260 = 1.62$ with 47 degrees of freedom. A two-sided p-value is about 0.11, so the fixed 5% rule does not reject zero.

5. CONFIDENCE INTERVAL

Using critical value near 2.01, the 95% interval is 0.42% $\pm$ 0.52%, or roughly [-0.10%, 0.94%]. Both harm and strong value remain compatible.

6. PRACTICAL THRESHOLD

The interval contains the 0.15% hurdle but its lower bound is below it. The study cannot yet establish that minimum useful net return is exceeded.

7. DEPENDENCE STRESS

If overlapping holdings raise the robust SE to 0.34%, t falls to 1.24 and the interval widens. Point estimate is unchanged, but independent evidence was overstated.

8. DECISION RECORD

Do not declare success or no effect. Continue only if new independent months, a paired variance reduction, or a cheaper implementation can resolve the economically relevant uncertainty.

Failure modes and diagnostic actions

INTERMEDIATE

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Tiny p, tiny net effect	Huge n; immaterial edge	Interval vs hurdle	Reject deployment claim
t falls with HAC	Serial dependence	ACF; lag sensitivity	Use robust uncertainty
Only best model works	Search selection	Full variant ledger	Correct and confirm
Subgroup signs differ	Thin or noisy states	Interaction interval	Pool or collect data
Bootstrap too narrow	Rows resampled IID	Block diagnostics	Resample valid units
Holdout improves repeatedly	Adaptive peeking	Access and change log	Retire holdout
Hit rate high; P&L low	Asymmetric payoff	Gain/loss and tails	Model economics
Significance vanishes live	Drift or leakage	Definition and timing diff	Rollback; replay

TRIAGE ORDER

Confirm population, timing, estimand, and data integrity first. Then inspect independent units, uncertainty, search family, effect size, and only afterward the threshold decision.

Final active recall and system index

INTERMEDIATE

QUESTION DISCIPLINE

State action, population, estimand, null, alternative, side, minimum useful effect, and error costs before confirmatory analysis.

INFORMATION STRUCTURE

Name sampling unit, clusters, overlap, time dependence, weights, missingness, regime assumptions, and effective information.

MAGNITUDE FIRST

Report raw and standardized effects, intervals, cost stress, contribution, and practical hurdle before threshold language.

CALIBRATION

Choose analytic, HAC, cluster, permutation, or bootstrap uncertainty from the null mechanism and data topology.

SEARCH CONTROL

Preserve the full family, correct discovery evidence, lock specifications, retire consumed holdouts, and confirm selected pipelines independently.

INTERPRETATION

P is not the probability the null is true; non-rejection is not equivalence; significance in one subgroup alone is not an interaction.

ROBUSTNESS

Inspect influence, tails, temporal stability, placebo results, alternative specifications, block choices, and method disagreement.

DECISION SYSTEM

Use tiered gates that combine adjusted evidence, lower bound versus hurdle, stability, costs, capacity, lineage, owner, and rollback.

PRODUCTION CONTINUITY

Monitor the same metric definition that earned approval, record protocol changes, preserve historical artifacts, and expire stale evidence.